S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 12

Aim: To design and develop an AI-based Customer Support Chatbot for an e-commerce platform that can assist users with order tracking, return policies, and product-related queries.

Name of Student: Prathamesh Deshmukh

Roll No.: CS23115

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AIM: To design and develop an AI-based Customer Support Chatbot for an e-commerce platform that can assist users with order tracking, return policies, and product-related queries.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

- To understand the basic concept of chatbot systems
- To learn how AI/NLP is used in customer support
- To implement a chatbot using simple programming techniques
- To handle user queries and generate appropriate responses
- To improve problem-solving and logical thinking skills
- To understand real-world applications of AI in e-commerce

Working:

The chatbot works by taking user input in the form of text and processing it to identify the user's intent.

- The system checks keywords such as "order", "return", or "product"
- Based on the detected intent, it selects a suitable response
- If the query is about order status, it may ask for an Order ID
- The chatbot then retrieves data (from predefined dataset or logic)
- Finally, it generates a response and displays it to the user

This chatbot can be:

- Rule-based (if-else conditions)
- OR enhanced using Natural Language Processing (NLP)

Technologies Used:

- Programming Language: Python
- Frontend (optional): HTML, CSS, JavaScript
- Framework: Flask / Streamlit
- Libraries:
    - NLTK or simple string matching
    - JSON (for storing product/order data)
- Tools: VS Code / Google Colab

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="AI-Powered Customer Support Chatbot for e-commerce — track orders, learn about return policies, and get product information instantly.">
  <title>ShopAssist AI — Customer Support Chatbot</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap" rel="stylesheet">
  <link rel="stylesheet" href="index.css">
</head>
<body>
  <!-- Animated Background -->
  <div class="bg-animation">
    <div class="bg-orb bg-orb-1"></div>
    <div class="bg-orb bg-orb-2"></div>
    <div class="bg-orb bg-orb-3"></div>
  </div>

  <div class="app-container">
    <!-- Header -->
    <header class="chat-header" id="chat-header">
      <div class="header-left">
        <div class="bot-avatar" id="bot-avatar">
          <svg width="28" height="28" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="1.8" stroke-linecap="round" stroke-linejoin="round">
            <path d="M12 2a4 4 0 0 1 4 4v1h1a3 3 0 0 1 3 3v8a3 3 0 0 1-3 3H7a3 3 0 0 1-3-3v-8a3 3
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0 0 1 3-3h1V6a4 4 0 0 1 4-4z"/>
                <circle cx="9" cy="13" r="1.2" fill="currentColor"/>
                <circle cx="15" cy="13" r="1.2" fill="currentColor"/>
                <path d="M9 17c1 1 5 1 6 0"/>
              </svg>
              <span class="status-dot"></span>
            </div>
            <div class="header-info">
              <h1 id="header-title">ShopAssist AI</h1>
              <span class="header-status" id="header-status">Online • Ready to help</span>
            </div>
          </div>
          <div class="header-right">
            <button class="header-btn" id="btn-clear" title="Clear Chat" aria-label="Clear chat history">
              <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor"
stroke-width="2" stroke-linecap="round" stroke-linejoin="round">
                <polyline points="3 6 5 6 21 6"></polyline>
                <path d="M19 6v14a2 2 0 01-2 2H7a2 2 0 01-2-2V6m3 0V4a2 2 0 012-2h4a2 2 0 012
2v2"></path>
              </svg>
            </button>
            <button class="header-btn" id="btn-theme" title="Toggle Theme" aria-label="Toggle
light/dark theme">
              <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor"
stroke-width="2" stroke-linecap="round" stroke-linejoin="round" class="icon-sun">
                <circle cx="12" cy="12" r="5"></circle>
                <line x1="12" y1="1" x2="12" y2="3"></line>
                <line x1="12" y1="21" x2="12" y2="23"></line>
                <line x1="4.22" y1="4.22" x2="5.64" y2="5.64"></line>
                <line x1="18.36" y1="18.36" x2="19.78" y2="19.78"></line>
                <line x1="1" y1="12" x2="3" y2="12"></line>
                <line x1="21" y1="12" x2="23" y2="12"></line>
                <line x1="4.22" y1="19.78" x2="5.64" y2="18.36"></line>
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                <line x1="18.36" y1="5.64" x2="19.78" y2="4.22"></line>
            </svg>
            <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round" class="icon-moon">
                <path d="M21 12.79A9 9 0 1111.21 3a7 7 0 009.79 9.79z"></path>
            </svg>
        </button>
    </div>
</header>

<!-- Chat Area -->
<main class="chat-area" id="chat-area">
    <!-- Welcome Message -->
    <div class="welcome-container" id="welcome-container">
        <div class="welcome-icon">
            <svg width="48" height="48" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="1.5" stroke-linecap="round" stroke-linejoin="round">
                <path d="M21 15a2 2 0 01-2 2H7l-4 4V5a2 2 0 012-2h14a2 2 0 012 2z"/>
            </svg>
        </div>
        <h2>Welcome to ShopAssist AI</h2>
        <p>Your intelligent shopping companion. I can help you with order tracking, returns, and product information.</p>
        <div class="quick-actions" id="quick-actions">
            <button class="quick-btn" data-query="Where is my order?">
                <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><path d="M21 16V8a2 2 0 00-1-1.73l-7-4a2 2 0 00-2 0l-7 4A2 2 0 003 8v8a2 2 0 001 1.73l7 4a2 2 0 002 0l7-4A2 2 0 0021 16z"/><polyline points="3.27 6.96 12 12.01 20.73 6.96"/><line x1="12" y1="22.08" x2="12" y2="12"/></svg>
                Track Order
            </button>
            <button class="quick-btn" data-query="What is your return policy?">
                <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor"
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stroke-width="2"><polyline points="1 4 1 10 7 10"/><path d="M3.51 15a9 9 0 102.13-9.36L1 10"/></svg>
                Return Policy
            </button>
            <button class="quick-btn" data-query="Tell me about iPhone 14">
                <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><rect x="5" y="2" width="14" height="20" rx="2" ry="2"/><line x1="12" y1="18" x2="12.01" y2="18"/></svg>
                Product Info
            </button>
            <button class="quick-btn" data-query="I need help with a refund">
                <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><line x1="12" y1="1" x2="12" y2="23"/><path d="M17 5H9.5a3.5 3.5 0 000 7h5a3.5 3.5 0 010 7H6"/></svg>
                Refund Help
            </button>
        </div>
    </div>

    <!-- Messages Container -->
    <div class="messages-container" id="messages-container"></div>

    <!-- Typing Indicator -->
    <div class="typing-indicator" id="typing-indicator">
        <div class="typing-avatar">
            <svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2">
                <path d="M12 2a4 4 0 0 1 4 4v1h1a3 3 0 0 1 3 3v8a3 3 0 0 1-3 3H7a3 3 0 0 1-3-3v-8a3 3 0 0 1 3-3h1V6a4 4 0 0 1 4-4z"/>
            </svg>
        </div>
        <div class="typing-dots">
            <span></span><span></span><span></span>
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            </div>
        </div>
    </main>

    <!-- Input Area -->
    <footer class="chat-input-area" id="chat-input-area">
        <div class="input-wrapper">
            <textarea
                id="user-input"
                placeholder="Type your message..."
                rows="1"
                aria-label="Type your message"
                autocomplete="off"
            ></textarea>
            <button class="send-btn" id="send-btn" title="Send Message" aria-label="Send message" disabled>
                <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round">
                    <line x1="22" y1="2" x2="11" y2="13"></line>
                    <polygon points="22 2 15 22 11 13 2 9 22 2"></polygon>
                </svg>
            </button>
        </div>
        <span class="input-hint">Press Enter to send • Shift+Enter for new line</span>
    </footer>
  </div>

  <script src="script.js"></script>
</body>
</html>
```
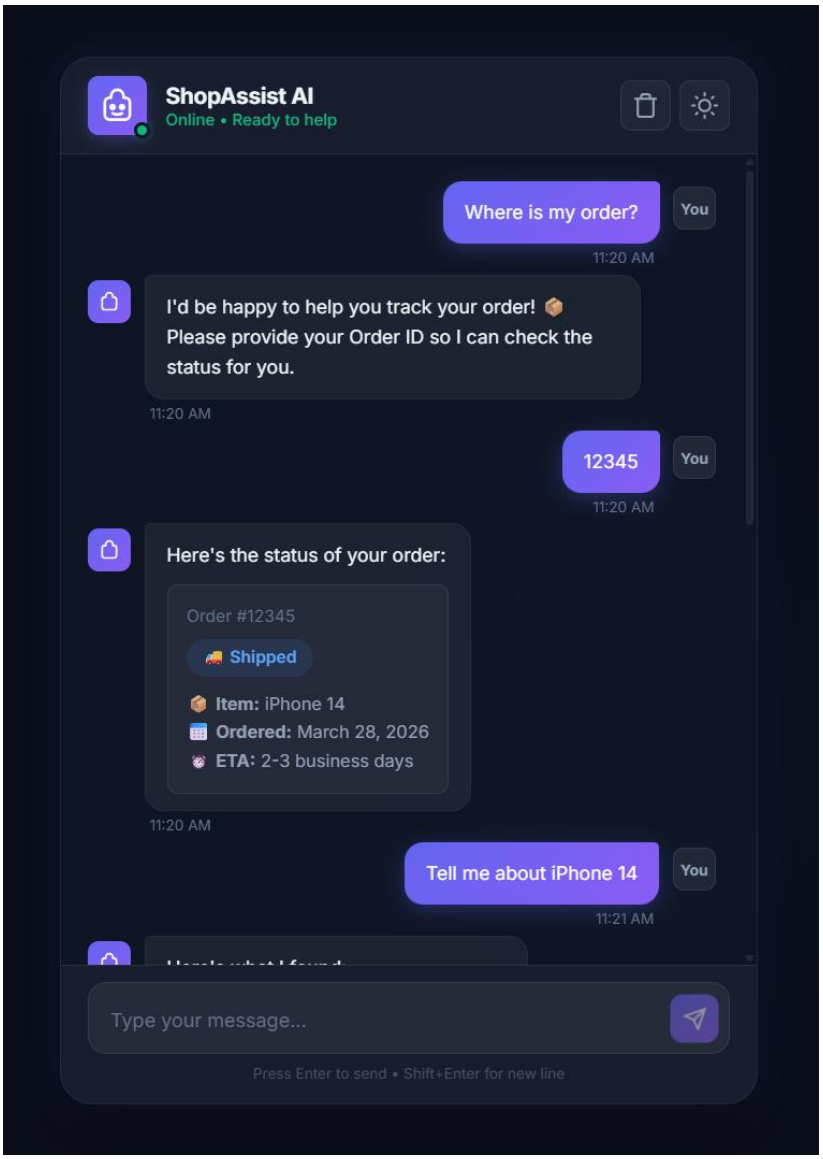
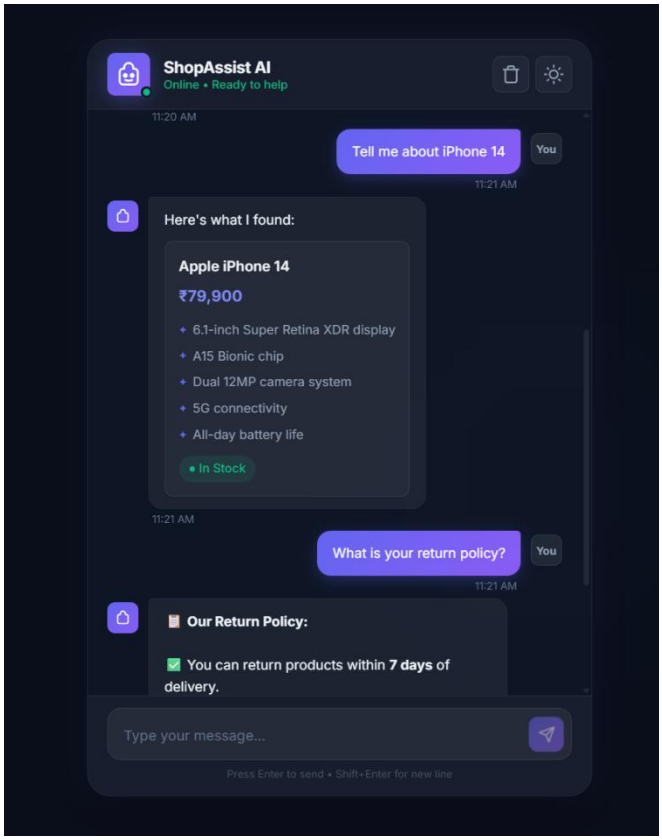
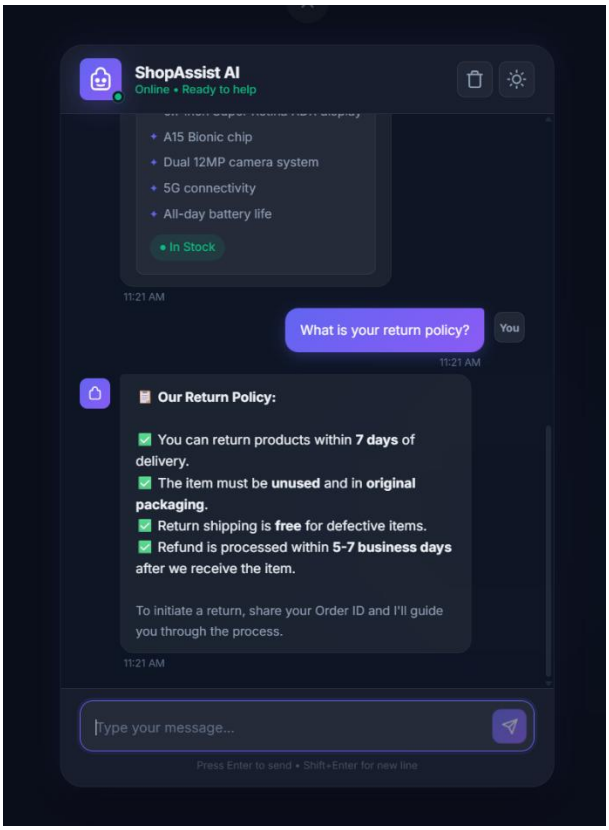

SCREENSHOTS:







Conclusion

Hence, the Customer Support Chatbot for an e-commerce platform was successfully implemented. The system is able to handle user queries related to order status, return policy, and product information effectively. This practical helped in understanding the working of chatbots and the application of AI in real-world customer support systems.

References

- W3Schools – https://www.w3schools.com
- GeeksforGeeks – https://www.geeksforgeeks.org
- Mozilla Developer Network – https://developer.mozilla.org